MILITARY SPECIFICATION

COUPLERS, DIRECTIONAL (COAXIAL LINE OR WAVEGUIDE), GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for directional couplers (see 6.3) for use with coaxial lines or waveguides. The couplers specified herein are used for Armed-Services application in the transmission lines of radar and radio equipment to inject or sample, at attenuated levels, the radio-frequency energy transmitted therein, and are suitable for operation over a temperature range of -65° to +95°C. (See 6.1.)

1.2 Military part number classification. The military part number shall consist of the letter "M", the basic number of the specification sheet, and an assigned dash number (see 3.1), as shown in the following example:

M15370/3 - 001

Military designator and specification sheet number — Dash number

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-M-44	-	Magnesium Alloy Plate and Sheet (AZ31B).
QQ-S-365	-	Silver Plating, Electrodeposited; General Requirements For.
TT-P-645	-	Primer, Paint, Zinc-Chromate, Alkyd Type.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

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MIL-W-85	-	Waveguides, Rigid, Rectangular, General Specification For.
MIL-P-116	-	Preservation, Methods of.
MIL-M-3171	-	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on.
MIL-L-3890	-	Lines, Radio Frequency Transmission (Coaxial, Air Dielectric), General Specification For.
MIL-F-3922	-	Flanges, Waveguide, General Purpose, General Specification For.

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MIL-C-5541 - Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-C-39012 - Connectors, Coaxial, Radiofrequency; General Specification For.
MIL-P-45209 - Palladium Plating, Electrodeposited.
MIL-C-45662 - Calibration System Requirements.

(See supplement 1 for list of associated specification sheets.)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.

DRAWINGS

DEPARTMENT OF THE NAVY

REA 49330 - UG-45/U Connector for Use With 7/8 Coaxial Air Dielectric Line.
REA 49331 - UG-46/U Connector for Use With 7/8 Coaxial Air Dielectric Line.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF DEFENSE

Handbook H4-1 - Federal Supply Code for Manufacturers (Part I).

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. Directional couplers furnished under this specification shall be products which have been tested and have passed the first article inspection specified in 4.4 (see 6.4).

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3.3 Materials. Materials shall be as specified herein; however, when a definite material is not specified, a material shall be used which will enable the directional couplers to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Copper alloy. Copper alloy used in the fabrication of waveguide-type directional couplers shall conform to the requirements for copper alloy specified in MIL-W-85. Copper alloy used in the fabrication of coaxial-type couplers shall conform to the material requirements specified in MIL-L-3890.

3.3.2 Aluminum alloy. Aluminum alloy used in the fabrication of waveguide-type directional couplers shall conform to the requirements for aluminum alloy specified in MIL-W-85.

3.3.3 Magnesium alloy. Magnesium alloy shall be composition AZ31B, condition H24, conforming to QQ-M-44. Unless otherwise specified (see 6.2), magnesium couplers shall not be supplied for naval applications.

3.3.4 Rubber. Rubber shall be suitable for use over the temperature range of -65° to +95°C.

3.4 Design and construction. Directional couplers shall be designed and constructed to conform to the requirements specified (see 3.1), in a manner entirely suitable for their intended use. They shall be of the lightest practicable weight consistent with the strength required for sturdiness, safety, and reliability. Couplers shall be fully shielded to radio-frequency fields. The frequency range shall be as specified (see 3.1).

3.4.1 Flanges. Flanges shall be so designed and manufactured as to provide the mating characteristics of the flange specified (see 3.1) in accordance with MIL-F-3922.

3.4.2 Connectors. Primary-line and secondary-line connectors shall be as specified (see 3.1). The material and gaging for receptacle connectors shall conform to the requirements of MIL-C-39012 or Drawing REA 49330 or REA 49331, whichever is applicable.

3.4.3 Caps. When required (see 6.2), weatherproof caps shall be attached to the directional coupler by means of small, flexible, corrosion-resistant chains.

3.4.4 Dimensions. The dimensions of the directional couplers shall be as specified (see 3.1).

3.4.5 Plating. All metal parts of directional couplers which are not corrosion-resistant shall have a plating of silver, type I or II, grade A, or palladium, in accordance with QQ-S-365 or MIL-P-45209, whichever is applicable. Aluminum-alloy surfaces shall be chemically treated in accordance with MIL-C-5541. Magnesium-alloy surfaces shall be given a chrome-pickle treatment conforming to type I of MIL-M-3171.

3.4.6 External finish. External finish shall be applied to the directional couplers regardless of plating or chemical treatment, except that the mating surfaces shall not be coated. The primer coat shall be zinc chromate conforming to TT-P-645. Two finish coats of enamel, conforming to type III, class 2, of MIL-E-15090, shall be applied. External coatings shall be applied as continuous films.

3.4.7 Screw threads. Screw threads shall be in accordance with Handbook H28.

3.5 Effective directivity. When directional couplers are tested as specified in 4.6.2, the effective directivity (in decibels (dB)) over the specified frequency range shall be the minimum specified (see 3.1).

3.6 Coupling. When directional couplers are tested as specified in 4.6.3, the measured value of coupling (in dB) at mid frequency (see 6.3.4) of the specified frequency range shall be as specified (see 3.1).

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3.7 Coupling variation. When directional couplers are tested as specified in 4.6.4, the variation in coupling (in dB) over the specified frequency range shall not exceed the maximum specified (see 3.1).

3.8 VSWR (voltage standing wave ratio).

3.8.1 Primary line (see 6.3.2). When directional couplers are tested as specified in 4.6.5.1, the maximum VSWR over the specified frequency range shall be as specified (see 3.1).

3.8.2 Secondary line (see 6.3.3). When directional couplers are tested as specified in 4.6.5.2, the maximum VSWR over the specified frequency range shall be as specified (see 3.1).

3.9 Insertion loss (when applicable, see 3.1). When directional couplers are tested as specified in 4.6.6, the insertion loss (see 6.3.5) shall not exceed the value specified (see 3.1).

3.10 Power dissipation of secondary-line termination. When directional couplers are tested as specified in 4.6.7, the VSWR of the secondary line at mid frequency of the specified frequency range (see 3.1) shall be as specified in 3.8.2.

3.11 Pressurization. When directional couplers are tested as specified in 4.6.8, there shall be no leakage, as detected by the continuous formation of escaping air bubbles.

3.12 Temperature cycling. When directional couplers are tested as specified in 4.6.9, there shall be no evidence of damage.

3.13 Moisture resistance. When directional couplers are tested as specified in 4.6.10, there shall be no change in electrical performance or destructive corrosion. Destructive corrosion shall be construed as any type of corrosion which in any way interferes with mechanical performance or appearance.

3.14 Vibration, high frequency. When directional couplers are tested as specified in 4.6.11, there shall be no change in electrical or mechanical performance.

3.15 Salt spray (corrosion). When directional couplers are tested as specified in 4.6.12, there shall be no destructive corrosion or pitting. Destructive corrosion shall be construed as any type of corrosion which in any way interferes with mechanical or electrical performance, or appearance.

3.16 Marking. Directional couplers shall be marked with the part number, the manufacturer's code, and the coupling value (in dB), in the location specified (see 3.1). Code-designating numbers shall be in accordance with Handbook H4-1. Marking shall be applied directly on the coupler, or on an attached nameplate. On unidirectional couplers, an arrow (or other indication) with the letters "ANT" shall be placed so as to point in the direction of the antenna, or in the direction in which incident power flows. On bidirectional couplers which have two different couplings, the coupling of each secondary line shall be indicated separately in a manner that will make clear which coupling is associated with each secondary line. Bidirectional couplers shall be marked to indicate which of the secondary lines has a nominal response to each direction of incident power flow. On bidirectional, special, or laboratory types, other appropriate marking may be substituted as specified (see 6.2).

3.17 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.1.1.1 Accuracy of test equipment. The frequency-measuring device used shall have an accuracy of ±0.1 percent, or better. The overall accuracy in the determination of VSWR shall be better than ±2 percent. The coupling-measuring system used shall have an accuracy of ±0.1 dB up to 10 dB and ±0.1 dB per 10 dB above 10 dB, or better.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.4).

(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all measurements and tests shall be made at room ambient temperature, pressure, and humidity.

4.4 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.4.1 Sample size. Two couplers of the same type shall be subjected to first article inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown.

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Test paragraph
Visual and mechanical examination	3.1, 3.3 to 3.4.7 incl, 3.16, and 3.17	4.6.1
Effective directivity	3.5	4.6.2
Coupling	3.6	4.6.3
Coupling variation	3.7	4.6.4
VSWR of primary line	3.8.1	4.6.5.1
VSWR of secondary line	3.8.2	4.6.5.2
Insertion loss (when applicable)	3.9	4.6.6
Power dissipation of secondary-line termination	3.10	4.6.7
Pressurization	3.11	4.6.8
Temperature cycling	3.12	4.6.9
Moisture resistance	3.13	4.6.10
Vibration, high frequency	3.14	4.6.11
Salt spray (corrosion)	3.15	4.6.12

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4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of all couplers of one type produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examination and tests specified in table II, and shall be performed in the order shown.

TABLE II. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination - - - - -	3.1, 3.3 to 3.4.7 incl, 3.16, and 3.17	4.6.1	1.0	4.0
Coupling - - - - - - - - - - - - - - - - - - -	3.6	4.6.3	1.0	---
Coupling variation - - - - - - - - - - - - - -	3.7	4.6.4	1.0	---

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table II. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table III, and the sample shall be selected from inspection lots that have passed the group A inspection.

TABLE III. Group B inspection.

Test	Requirement paragraph	Test paragraph
Effective directivity - - - - - - - - - - -	3.5	4.6.2
VSWR- - - - - - - - - - - - - - - - - - - -	3.8	4.6.5

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The AQL shall be 6.5 percent defective.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order if the lot is accepted and the sample units are still within specified electrical tolerances.

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4.5.2 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preparation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.6 Test procedures.

4.6.1 Visual and mechanical examination. Directional couplers shall be inspected to verify that the materials, design and construction, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3 to 3.4.7 inclusive, 3.16, and 3.17).

4.6.2 Effective directivity (see 3.5). The effective directivity of directional couplers shall be determined over the specified frequency range (see 3.1). The effective directivity of unidirectional couplers shall be computed from the ratio, taken as greater than unity and expressed in dB, of the available powers at the output of the secondary line for the two directions of excitation, at equal power levels, of the primary line; the secondary line shall be terminated in a matched detector. The effective directivity of bidirectional couplers shall be determined separately for each secondary line. At the option of the Government (see 6.2), the alternate test specified in 4.6.2.1 may be used.

4.6.2.1 Alternate test. The effective directivity of directional couplers shall be determined over the specified frequency range. The effective directivity shall be computed from the difference between measurements of minimum and maximum outputs of a secondary line, when a short circuit is displaced through one-half a wave-length in the primary line. The generator at the primary line shall be matched to the primary line. The effective directivity of bidirectional couplers shall be determined separately for each secondary line with the remaining secondary lines terminated by an impedance equal to the characteristic impedance of the secondary line.

4.6.3 Coupling (see 3.6). The coupling of directional couplers shall be determined at mid frequency in the specified frequency range (see 3.1). The coupling of unidirectional couplers shall be determined as the ratio, expressed in dB, of the power input to the primary line to the power available at the output of the secondary line, with the output end of the primary line properly terminated, when the incident power is in the direction of excitation of the primary line that will produce the maximum response in the secondary line. The coupling of bidirectional couplers shall be determined separately for each secondary line.

4.6.4 Coupling variation (see 3.7). The coupling variation of directional couplers shall be determined over the specified frequency range (see 3.1). The coupling variation of unidirectional couplers shall be determined by taking the difference between the maximum and the minimum coupling over the specified frequency range. Such difference shall be taken as a positive number or zero. The coupling variation of bidirectional couplers shall be determined separately for each secondary line.

4.6.5 VSWR (see 3.8).

4.6.5.1 Primary line (see 3.8.1). The VSWR of directional couplers shall be measured over the specified frequency range (see 3.1). The VSWR of unidirectional couplers shall be measured at the input end of the primary line, with the output end of the primary line and the secondary line terminated in matched loads. The VSWR of bidirectional couplers shall be measured at one end of the primary line, with the other end terminated in a matched load; the secondary lines shall be terminated in matched loads.

4.6.5.2 Secondary line (see 3.8.2). The VSWR of directional couplers shall be measured over the specified frequency range (see 3.1). The VSWR of unidirectional couplers shall be measured at the output of the secondary line with both ends of the primary line terminated in matched loads. The VSWR of each secondary line of bidirectional couplers shall be measured at the output of the secondary line with the other secondary line and both ends of the primary line terminated in matched loads.

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4.6.6 Insertion loss (when applicable, see 3.1) (see 3.9). The insertion loss shall be measured by a substitution technique using radio, audio, or intermediate frequency (RF, AF, or IF). An average of three test runs shall be used as a final result.

4.6.7 Power dissipation of secondary-line termination (see 3.10). The power dissipation of the secondary-line termination shall be determined over the specified frequency range (see 3.1). The power (Pi) to be applied shall be determined from the following formula:

$$Pi = \frac{Po}{Log_{10}{}^{-1}\left(\frac{d}{10}\right)}$$

Where:

Pi = power.
Po = continuous-wave power rating of primary line (see 6.2).
d = coupling in dB.

The power (Pi) shall be applied for a period of 1 hour to the output of the secondary line, with both ends of the primary line terminated in matched loads. Within 3 minutes after the 1-hour period, the VSWR of the secondary line shall be measured as specified in 4.6.5.2, at mid frequency in the specified frequency range. At the option of the Government, the alternate test specified in 4.6.7.1 may be used.

4.6.7.1 Alternate test. The power dissipation of the secondary-line termination shall be determined over the specified frequency range. Rated continuous-wave power (see 6.2) shall be applied for a period of 1 hour to the input end of the primary line, with the output end of the primary line and the output of the secondary line terminated in matched loads. After the 1-hour period, the VSWR of the secondary line shall be measured as specified in 4.6.5.2, at mid frequency in the specified frequency range (see 3.1).

4.6.8 Pressurization (see 3.11). Directional couplers shall be subjected to an internal gas pressure of 30 pounds per square inch for at least 20 seconds while immersed in water at approximately 20°C. The ends of the primary line shall be appropriately sealed.

4.6.9 Temperature cycling (see 3.12). Directional couplers shall be tested in accordance with method 102, test condition D, of MIL-STD-202, except that the low temperature shall be -65°C and the high temperature shall be +95°C.

4.6.10 Moisture resistance (see 3.13). Directional couplers shall be tested in accordance with method 106 of MIL-STD-202. Vibration, step 7B, shall be performed.

4.6.11 Vibration, high frequency (see 3.14). Directional couplers shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting of specimens - Couplers shall be attached to the vibration table by means of clamps. A clamp shall be placed around the center of each coupler, and no part of the coupler shall touch any object other than the clamp.

(b) Test condition - B.

(c) Measurements - VSWR shall be measured as specified in 4.6.5.

4.6.12 Salt spray (corrosion) (see 3.15). Directional couplers shall be tested in accordance with method 101, test condition A, of MIL-STD-202. The RF ports of coaxial types shall not be sealed for this test. Waveguide types shall be sealed at the flanges during the salt spray test. After this test, the couplers shall be washed, dried, and examined for destructive corrosion or pitting.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Directional couplers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Directional couplers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), directional couplers shall be individually packaged in accordance with MIL-P-116, submethod IIe insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. The container shall conform to PPP-B-566, PPP-B-676 or PPP-B-636.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Cleaned and dried directional couplers shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged directional couplers shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

5.2.2 Level B. The packaged directional couplers shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto. For Army procurement, fiberboard containers shall be class weather-resistant as specified in level A.

5.2.3 Level C. The packaged directional couplers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Directional couplers, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

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5.2.4.2 Level B. Directional couplers, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic. For Army procurement, caps shall be weather resistant as specified in level A.

5.2.4.3 Level C. Directional couplers, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Intended use. Directional couplers may be used in conjunction with a radio-frequency source for injecting a radio-frequency wave into a transmission line so that it flows in one direction only. Directional couplers may also be used to sample a radio-frequency wave flowing in a particular direction in a transmission line while accepting relatively insignificant portions of a radio-frequency wave flowing in the opposite direction.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of applicable specification sheet and complete part number (see 3.1).

(c) For naval applications, whether magnesium may be used in fabricating directional couplers (see 3.3.3).

(d) Whether weatherproof caps are required (see 3.4.3).

(e) Special marking required (see 3.16).

(f) Specify when alternate test for effective directivity is required (see 4.6.2).

(g) Continuous wave power rating of primary line (see 4.6.7).

(h) Levels of preservation and packaging, packing, and applicable marking (see section 5).

6.3 Definitions. For the purpose of this specification, the following definitions shall apply:

6.3.1 Directional coupler. A directional coupler is a transmission-line component characterized physically by two (or three, for certain bidirectional couplers) juxtaposed transmission lines and an associated coupling structure through which a transfer of RF energy from one line to the other is effected; its electrical behavior is characterized ideally by such interaction between the two lines that excitation in a single direction in either line produces a response in the companion line in one direction only.

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6.3.1.1 Unidirectional coupler. A unidirectional coupler is a directional coupler so designed as to provide a nominal response in the secondary line to propagation in the primary line, in one direction only.

6.3.1.2 Bidirectional coupler. A bidirectional coupler is a directional coupler so designed as to provide separate and simultaneous nominal responses in the secondary lines to each of the two directions of propagation in the primary line.

6.3.2 Primary line. The primary line of a directional coupler is the line designed to receive the principal flow of RF energy in the transmission line to which the directional coupler is adjunct. The input end of the primary line of a unidirectional coupler is the end into which power must flow in order to produce the maximum power at the output of the secondary line. The other end is called the output (antenna) end.

6.3.3 Secondary line. The secondary line of a directional coupler is the line that is coupled to the primary line by means of the coupling structure. In a unidirectional coupler there is one secondary line. In a bidirectional coupler the secondary lines have a nominal response to a different direction of propagation in the primary line. In some bidirectional couplers the secondary lines are physically separate; in others they run physically together.

6.3.4 Mid frequency. The mid frequency of a directional coupler is defined as the arithmetic mean of the limits of its frequency range.

6.3.5 Insertion loss. The loss produced by adding (inserting) a device into a signal transmission path.

6.4 First article. Invitations for bid should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI
Navy - OS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, WC
Navy - MC, CG, AS, SH
Air Force -

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0709)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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